



HYBRID TOUCHLESS & KEYPAD CITIZEN MANUAL

USER OPERATION MANUAL

صارفین اور شہریوں کیلئے معلوماتی رہنما

Comprehensive guide for public recycling, hybrid touchless QR scan and tactile keypad operation, multi-sensor deposit validation, instant digital rewards, and dual-screen kiosk displays.

**Hybrid Start**

Scan on-screen QR with RVM Mobile App or press Key 0 on tactile keypad.

**Idle Expansion**

50% screen instructional expansion attracting citizens with instant wake-up.

**Touchless Rewards**

Zero-contact mobile wallet sync via dynamic QR scan or 11-digit mobile entry.

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APPLICABILITY

RVMDesktopApp
Platform

KIOSK LOCATION & GPS

Katra Neem Wala, Lahore
(31.5826° N, 74.3276° E)

Manual Overview & Navigation

A structured roadmap of all user interactions, kiosk screen displays, and environmental reward steps.

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⚡ HYBRID RECYCLING FOR CITIZENS: TOUCHLESS OR KEYPAD Option 1 (100% Touchless): Open RVM Mobile App → Scan the on-screen 'SCAN TO START' QR Code → Safety gate opens automatically → Insert empty container → Press Enter or finish via app → Scan dynamic wallet QR code → Collect instant reward points! Option 2 (Tactile Keypad): Walk up to kiosk → Press '0' on keypad → Insert containers bottom-first → Wait for green checkmark → Press Enter → Type 11-digit mobile number → Rate your experience → Collect points!	

1. Reverse Vending Machine Overview

An intelligent, multi-material automated recovery unit engineered for university campuses and public plazas.

HYBRID CITIZEN TOUCHPOINTS

The RVM Kiosk combines tactile physical inputs with state-of-the-art mobile touchless scanning for maximum convenience and hygienic public safety:

- Touchless QR Code Interaction:** Citizens can authenticate and start recycling without touching the machine by scanning the dynamic on-screen QR code using their smartphone.
- Interactive Portrait Kiosk Display (1080x1920):** Primary touch interaction showing bilingual animations, live sensor results, and points balance.
- Digital Signage Display (Secondary Screen):** Loops continuous environmental public service announcements and partner sustainability media.

 **SUSTAINABILITY MISSION**

Every container deposited directly prevents landfill dumping: **+0.15 kg CO2** emission prevented and **+0.75 Liters** of clean water conserved per verified container.

KEY HARDWARE TOUCHPOINTS

Touchpoint Component	Citizen Function
Touchless QR Card	High-contrast on-screen QR badge for camera scan session initiation.
Illuminated Aperture	Circular entry slot with white LED guide ring. Accepts PET bottles, cans, and UBC cartons.
Motorized Safety Gate	Anti-pinch protective door opens automatically when QR is scanned or 0 is pressed.
Physical Matrix Keypad	12-key tactile keypad mounted at ergonomic height. Resistant to rain, dust, and sunlight.
Instructional Container	High-visibility animated display showing real-time feedback and celebration animations.
Reward Balance Display	Instant point tracker displaying accumulated session points in large bold digits.

CITIZEN SAFETY NOTICE

The intake chamber is monitored by safety sensors. Never attempt to force foreign objects, glass, or pressurized cans into the machine. The safety door will immediately halt if an obstruction is detected.

SUPPORTED RECYCLABLE MATERIALS

 **Plastic PET Bottles**

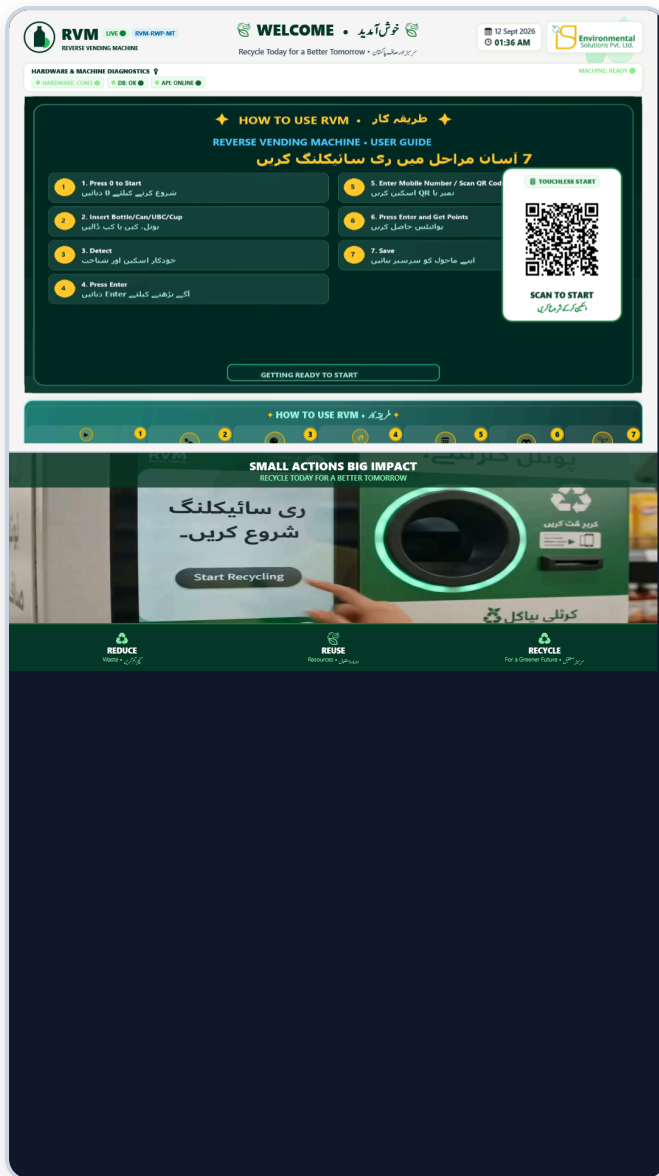
Water, soda, juice bottles (250ml to 2.25L). Must be empty of liquid. S, M, L accepted.

 **Aluminium Cans**

Beverage cans, soda tins. Verified by internal inductive sensor. S, M, L accepted.

 **UBC TetraPak Cartons**

Juice cartons, milk packs. Verified by ultrasonic acoustic bounce. S, M, L accepted.



STANDBY SCREEN • SCREEN 02

Home Page & Citizen Dashboard

مرکزی اسکرین اور طریقہ کار

When the RVM is in its ready standby state, it presents the primary welcome screen. The interface is meticulously balanced with high-contrast typography, emerald eco-card containers, bilingual Urdu and English text, and a prominent **Touchless Start QR Card**.

TWO WAYS TO BEGIN

- 1. Touchless:** Scan the on-screen 'TOUCHLESS SCAN TO START' QR code with the RVM Mobile App.
- 2. Keypad:** Press the '0' key on the physical keypad or tap the on-screen start button.

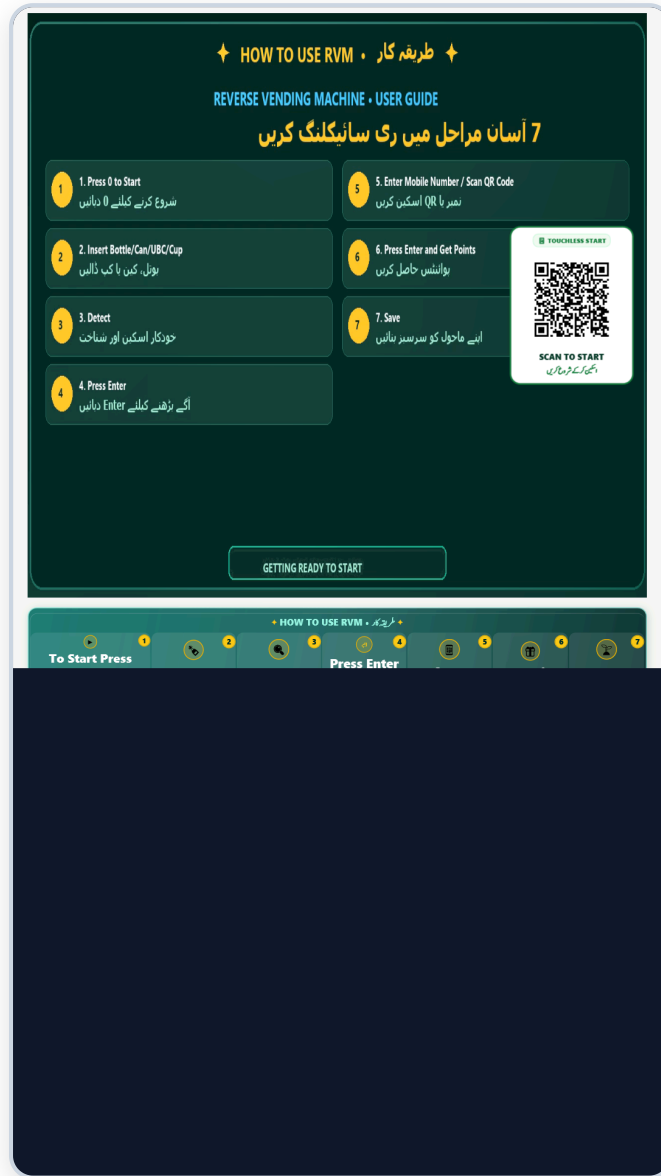
Dashboard Section	Citizen Feature & Information
Touchless QR Card	Glowing card on the centerpiece displaying dynamic QR code for zero-contact smartphone initiation.
Top Header Bar	Machine ID (RVM-2026), Real-time Live Status badge, and current Pakistan Standard Time.
Hardware Diagnostics	Green status dots confirming Serial connection (COM3), Local Database (OK), and Central Cloud API (ONLINE).
Instructional Player	16:9 video widget running bilingual instructional motion guides and material animations.
How to Use RVM Bar	Teal glassmorphism card outlining the 7 chronological steps of the recycling journey.
Session Breakdown	Live material tallies (Plastic, Cans, UBC, Rejected) and large Reward Balance banner.

🏆 COMMUNITY RECOGNITION

Top campus recyclers are ranked live on the kiosk system. Your username and reward points will be showcased to motivate the entire community!

2.2 Kiosk Idle Mode & Display Expansion

Automatic display optimization when the kiosk is inactive for more than 1 minute to attract pedestrians and showcase educational content.



STANDBY MODE - 60s INACTIVITY

1-Minute Inactivity Idle Screen

ایک منٹ بعد اسکرین کی خود کار توسیع

When no citizen interacts with the kiosk for over **1 minute (60 seconds)**, the application automatically transitions into cinematic standby mode while maintaining complete touchless readiness:

Layout Element	Behavior During Idle Mode
Instructional Video	Expands to 50% of screen height (960px) with smooth zoom-in styling inside container.
Touchless QR Access	Remains active and scanned by mobile app to trigger instant session wake-up.
How to Use RVM Card	Stays 100% visible with all 7 step cards and Urdu text clearly legible below the video.
Ad Video Section	Stays 100% visible & playing in the remaining lower screen area.
Zero Overlap Geometry	Strict Auto row stacking guarantees zero visual clipping or touching between cards.

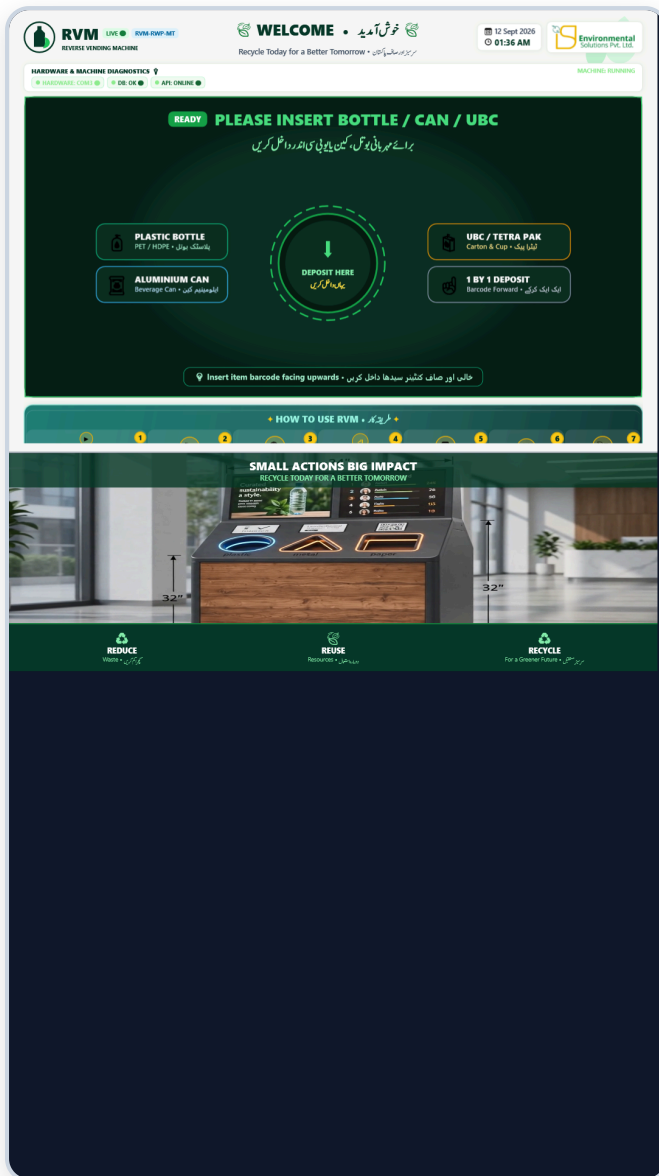
HOW TO RETURN & START RECYCLING

Scan the on-screen Touchless QR Code with your RVM Mobile App OR press Key '0' on the keypad. The kiosk instantly wakes up, exits idle mode, and opens the intake gate!



ECO-FRIENDLY STANDBY MODE

The 1-minute idle mode automatically expands instructional visuals to welcome new citizens while optimizing kiosk power efficiency and component longevity.



STEP 1 OF 7 • TOUCHLESS QR OR KEY '0'

Starting Your Recycling Session

پہلا مرحلہ: سیشن کا آغاز (کیو آر اسکین یا ٹن 0)

To start a recycling session, citizens have two flexible options: scan the **Touchless QR Code** with the RVM Mobile App for a 100% contactless experience, OR press '0' on the tactile hardware keypad. The motorized gate opens, chamber lights turn on, and deposit sensors arm ready for container entry.

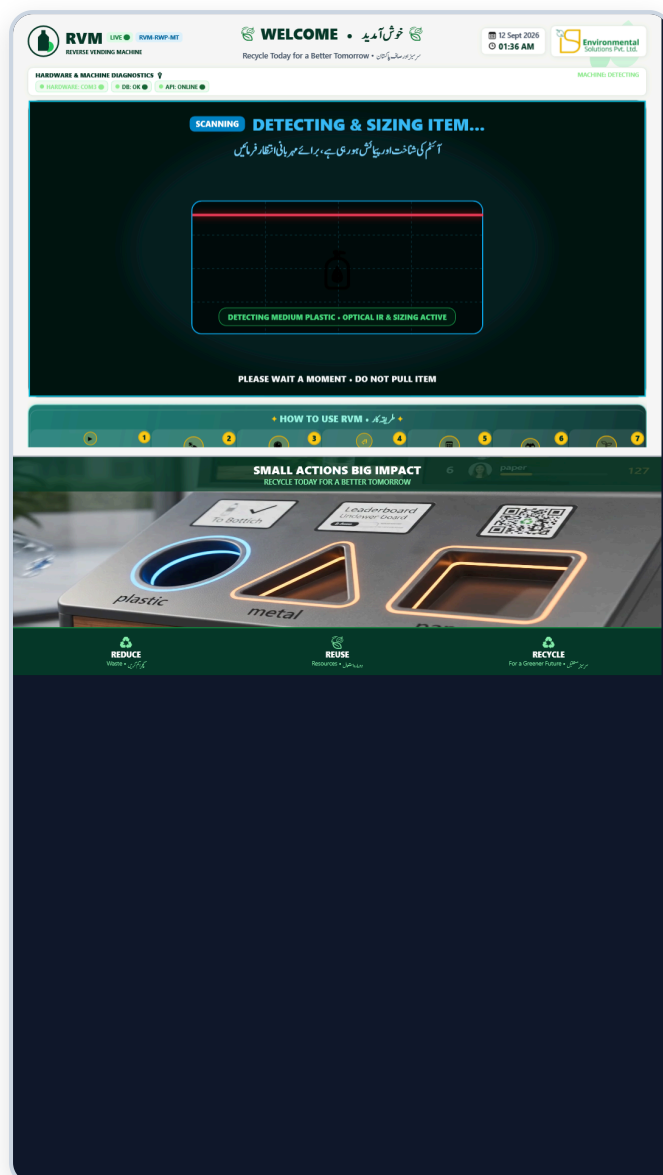
CITIZEN ACTION REQUIRED

Ensure container is empty of liquids. Insert bottle or can bottom-first into the illuminated circular aperture.

Mechanical & Electronic Event	Technical State
Entrance Door Servo	Rotates 90° to open position (Aperture Ready)
Chamber Lighting	High-intensity white LED ring illuminates interior
Acoustic Sensor Ping	Ultrasonic transducer arms (Pin D12/D13)
Safety Gate Sensor	Infrared safety beam active to prevent finger pinching
Session Timer	30-second activity countdown initiated

TIP FOR OPTIMAL RECYCLING

Always insert bottles bottom-first with the cap facing toward you. Keep barcodes visible if present to assist high-speed optical classification.



STEP 2 OF 7 • DETECTION & SIZING

Real-Time Sensor Scanning

دوسرا مرحلہ: اسکیننگ اور پیمائش

As the container settles inside the chamber, the RVM executes a sub-second multi-sensor sweep. The system cross-references optical height beams, inductive metallic response, and ultrasonic distance to classify material and calculate dimensions.

CITIZEN ACTION REQUIRED

Allow container to rest inside chamber. Wait 1 second while sensors verify dimensions and confirm acceptance.

Sensor Subsystem	Detection Criteria & Result
Inductive Sensor (D5)	Metal conductivity check (1 = Can / Metal, 0 = Plastic / UBC)
IR Optical Array (D2, D7, D8)	3-tier height measurement: Small (<16cm), Medium (16-24cm), Large (>24cm)
Ultrasonic Chamber (D12/D13)	Precise length measurement via high-frequency acoustic bounce
Liquid Detection	Weight & optical refraction test to detect residual liquids
Arbitration Time	Under 850 milliseconds for complete item classification

ACCEPTED SIZE CATEGORIES

Small: 250ml–350ml (+5 pts) • **Medium:** 500ml–1.0L (+10 pts) • **Large:** 1.5L–2.25L (+15 pts). All sizes are automatically awarded appropriate points!

Interactive Motion Overlays on Video Widget

During the recycling transaction, the video player displays rich dynamic overlays that guide the citizen through detection stages.

STATE 1

Please Insert Item

PLEASE INSERT BOTTLE / CAN

برائے مہربانی خالی بوتل یا کین ڈالیں

Displays an animated pulsating circular aperture showing bottles entering smoothly. Reassures citizens that the machine is armed and awaiting container deposit.

STATE 2

Detecting & Sizing

DETECTING & SIZING ITEM...

بوتل کی شناخت اور پیمائش ہو رہی ہے

Features a simulated bidirectional laser beam scanning vertically across the container. Confirms to the citizen that sensors are measuring material dimensions in real-time.

STATE 3

Accepted + Points

ITEM ACCEPTED! +10 PTS

بوتل قبول! پوائنٹس کا اندراج ہو گیا

Celebratory green notification with confetti graphics. Automatically plays material dancing animation (Plastic, Can, or TetraPak) matching the recognized item.

STATE 4

Item Rejected

ITEM REJECTED! PLEASE RETRIEVE

آئٹم نامنظور! برائے مہربانی واپس نکال لیں

Clear safety warning with gentle audio alert. Gate unlocks to allow retrieval of non-recyclable garbage, glass, or containers with remaining liquid.

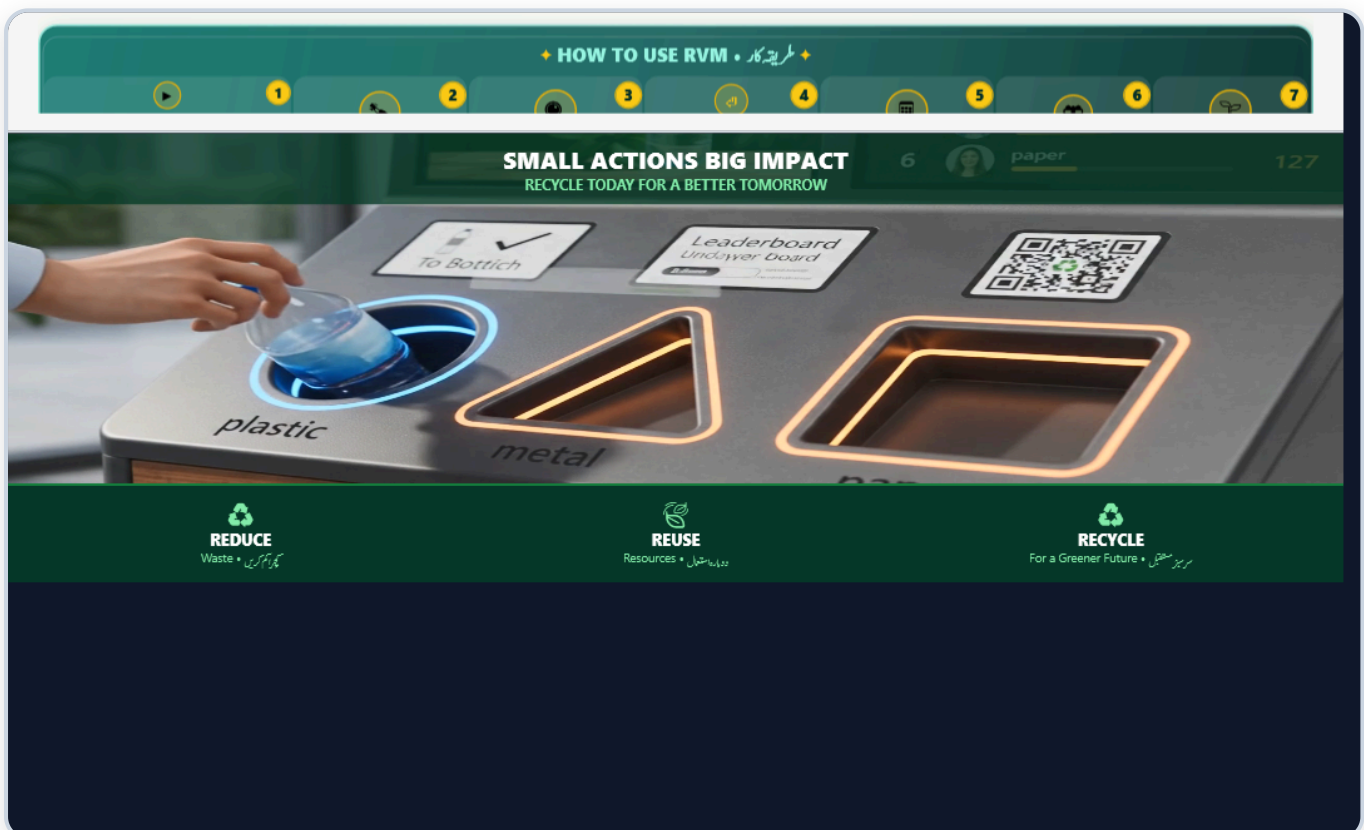
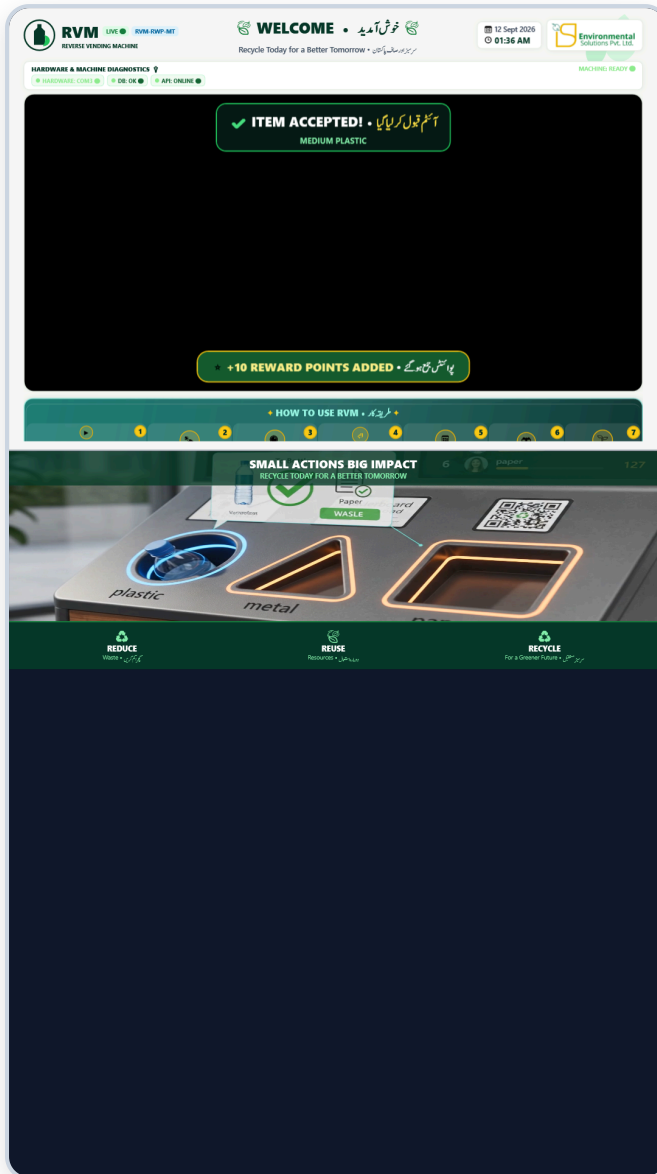


Figure 7.1: Real Kiosk Rejection Screen displaying safety alert and retrieval instructions.



STEP 4 OF 7 • ANTI-CHEAT VALIDATION

Drop Confirmation & Claiming Rewards

چوتھا مرحلہ: بوتل چیمبر میں گرنے کی تصدیق

To maintain reward integrity, points are only awarded after the anti-cheat drop gate actuates 180° and the ultrasonic sensor verifies the physical fall of the container into the internal bin. Once confirmed, the citizen can insert another item or press **Enter** to proceed to wallet credit.

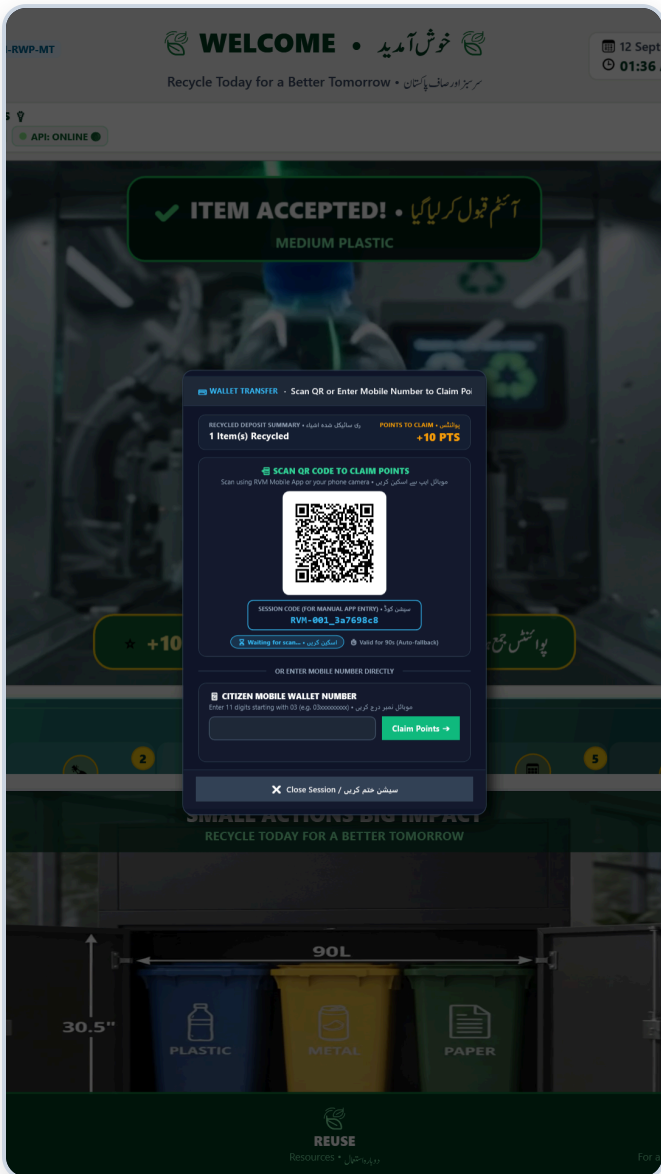
CITIZEN ACTION REQUIRED

Insert another container to earn more points, OR press 'Enter' on the keypad to finalize session and claim rewards.

Verification Parameter	Anti-Cheat Mechanism
Drop Gate Actuator	Servo Pin D8 rotates 180° downward to drop container into bin
Ultrasonic Fall Ping	Acoustic bounce detects item clearing chamber (BOTTLE:CLEARED)
String-Tie Anti-Cheat	If an item is pulled back via string, machine flags ERROR: CLEAR_TIMEOUT
Session Accumulation	Points and items increment live on dashboard breakdown

🎁 MULTI-CONTAINER SESSIONS

You do not need to enter your phone number for each bottle! Keep depositing containers one after another. When you are completely done, press **Enter** once to credit all accumulated points simultaneously.



STEPS 5 & 6 • TOUCHLESS QR OR MOBILE ENTRY

Claiming Points via QR Code Scan or Mobile Entry

پانچواں مرحلہ: کیو آر کوڈ اسکین یا موبائل نمبر سے انعام کا حصول

After pressing **Enter** to conclude container deposits, the kiosk displays the Points Claim modal. Citizens can claim points in two convenient ways: instantly scan the on-screen dynamic QR Code using the RVM Mobile App, or type their 11-digit mobile number using the physical keypad.

CITIZEN ACTION REQUIRED

Scan the on-screen QR Code using your RVM Mobile App (or phone camera), enter the 6-character Session Code in the app, OR type your 11-digit mobile number on the keypad and press 'Enter'.

Claim Method / Rule	Specification & Citizen Experience
Dynamic QR Code Scan	Instant, touchless wallet sync via RVM Mobile App or smartphone camera
Session Code (App Entry)	6-character code under QR for quick manual entry in the mobile app
Keypad Mobile Input	Enter 11-digit Pakistani mobile number (03001234567) + press 'Enter'
Backspace / Delete	Press '*' key on physical keypad to clear the last typed digit
Cancel Transaction	Press '#' key on keypad to cancel entry and return to deposit screen
90-Second Active Timer	Ensures session privacy and triggers safe automatic fallback if uncollected

POINTS REDEMPTION & WALLET BALANCE

Your points never expire! Rewards sync directly to your centralized account across all campus and public RVM kiosks, redeemable for mobile balance top-ups, cafeteria discounts, and gift cards.

STEP 7 OF 7 • CITIZEN EXPERIENCE RATING

Rating Your Recycling Experience

ساتواں مرحلہ: تجربے کی درجہ بندی اور رائے

Following point crediting, the kiosk presents a 1-to-5 star experience rating screen. Citizens can touch the stars to submit their rating and tap quick feedback tag chips ("Fast & Easy", "Clean Machine", "Great Rewards").

CITIZEN ACTION REQUIRED

Tap 1 to 5 stars on the screen or press a keypad digit (1-5), then press Enter to submit your feedback.

☆
HOW WAS YOUR EXPERIENCE?
آپ کا ری سائیکلنگ کا تجربہ کیسا رہا؟



★★★★★ 5/5 — Excellent • بہترین

Feedback Parameter	Citizen Feature
Star Rating Scale	1 to 5 Gold Stars (Poor to Excellent)
Feedback Tags	Fast & Easy, Clean Machine, Great Rewards, Needs Help
Auto-Skip Timeout	10-second timer automatically completes if citizen walks away
Final Receipt Screen	Displays total points earned and CO2 prevented
Auto-Reset	Machine cleanly resets session buffer for next user

☀️ CONTINUOUS IMPROVEMENT

Your feedback is transmitted directly to the University Administration and Municipal Waste Management dashboard to ensure the kiosk remains clean and well-maintained.

4. Dual-Screen Landscape Display Mode

Wide-screen interactive layout combining kiosk deposit controls (56%), live community leaderboard, and commercial signage video (44%).

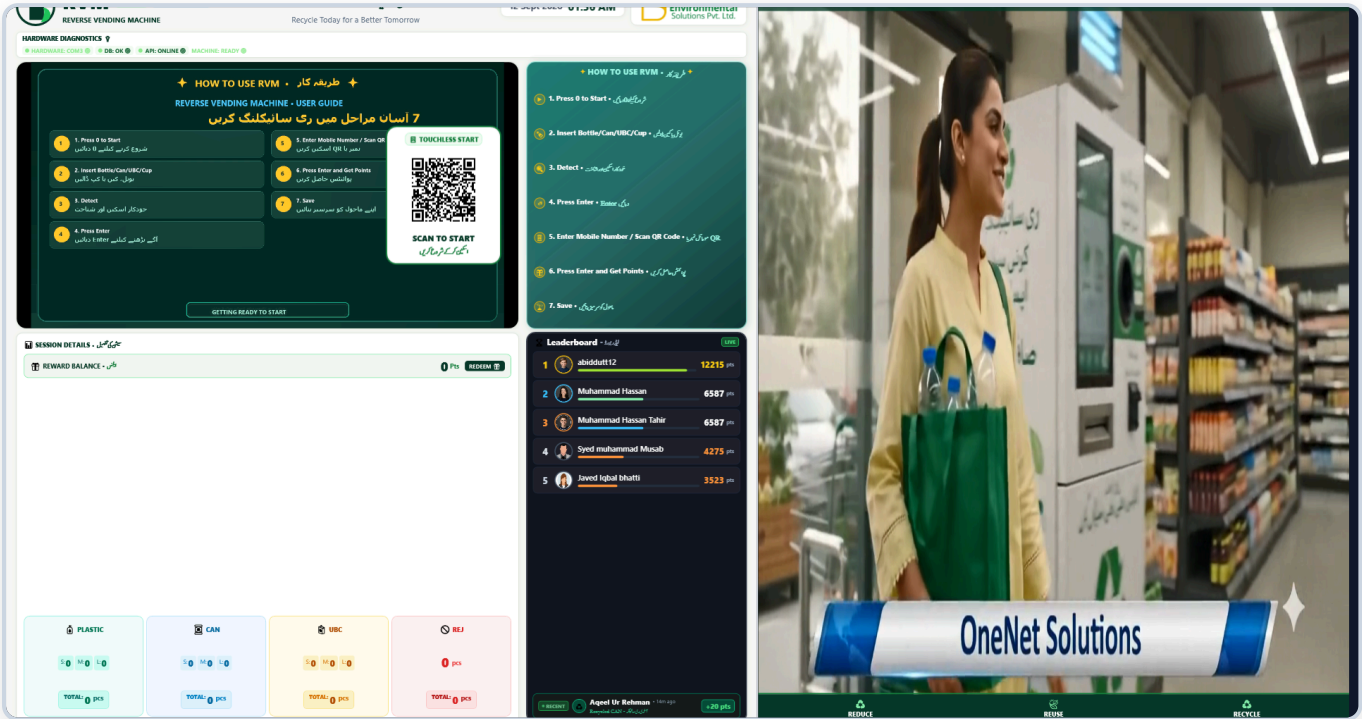


Figure 12.1: Secondary Public Kiosk Interface (1920×1080) running live in Metro Terminals and Shopping Atriums.

LEFT INTERACTION TIER (56% WIDTH)

Component	Function & Specification
Header Ribbon	Machine status badge, live clock, and partner branding card.
Diagnostics Strip	Green indicators for Serial COM3, SQL Database, and Cloud API.
How-To Centerpiece	Animated 7-step guide + Touchless QR code for mobile start.
Live Breakdown	Categorized counts for Plastic, Cans, UBC cartons, and Rejections.
Reward Points	Session points tally with instant Redeem voucher button.

SIGNAGE & LEADERBOARD (44% WIDTH)

Component	Function & Specification
Eco Champions	Dark arcade-style leaderboard ranking top 5 community recyclers.
Recycler Avatars	Custom visual identity avatars with dynamic progress bars.
Live Ticker	Recent activity ticker showcasing latest container deposits.
Signage Player	Loops 1080p commercial ads and environmental awareness videos.
Multi-Monitor Sync	Auto-routed to secondary HDMI LED screen via PowerShell launcher.



MULTI-MONITOR KIOSK DEPLOYMENT

In dual-monitor installations, Screen 0 (Primary 1080x1920) is dedicated to citizen touch interaction, while Screen 1 (Secondary 1920x1080) faces the pedestrian concourse for high-visibility public engagement and commercial revenue generation.

5. Hybrid Input & Keypad Quick Reference

Comprehensive guide to all tactile keypad inputs, touchless QR start triggers, and citizen commands.

Input Method	Screen State	Function & Kiosk Response
Touchless QR Scan	Standby / Idle Mode	Touchless Session Trigger: Scan on-screen 'SCAN TO START' QR code with RVM Mobile App. Opens door without touching machine!
Key 0	Standby / Idle Mode	Hardware Session Trigger: Opens safety entrance door, activates chamber lighting, and arms sensors. Exits idle video mode.
Enter ↵	Container Deposited	Finalize & Proceed: Concludes container intake and opens the points claim dialog.
Wallet QR Scan	Wallet Claim Screen	Touchless Points Claim: Scan the dynamic on-screen QR code with RVM Mobile App or phone camera for instant wallet crediting.
Session Code	Wallet Claim Screen	Manual Mobile App Entry: Type the 6-character session code shown below the QR code directly in the RVM Mobile App.
Digits 0–9	Wallet Claim Screen	Mobile Number Input: Types 11 digits for Pakistani mobile number (03xxxxxxxx) via physical keypad.
Enter ↵	Wallet Claim Screen	Submit Wallet: Validates 11-digit mobile number, credits points to database, and proceeds to rating screen.
Star Key (*)	Wallet Claim Screen	Backspace / Delete: Removes the last typed digit for instant correction.
Hash Key (#)	Any Modal / Dialog	Cancel / Back: Closes active dialog and returns to the previous screen.
Digits 1–5	Feedback Rating	Citizen Rating: Selects satisfaction level from 1 (Very Bad) to 5 (Excellent), followed by Enter.

HYBRID CITIZEN ACCESSIBILITY

Citizens can choose whichever method they prefer: 100% touchless mobile app interaction for hygiene, OR the tactile physical keypad for instant walk-up accessibility.

ERGONOMIC ACCESSIBILITY

The hardware keypad features embossed numbers and high-contrast tactile feedback compliant with universal accessibility standards for wheelchair users and visually impaired citizens.

6. Frequently Asked Questions & Troubleshooting

Common questions regarding recycling rewards, container acceptance, and citizen assistance.

Q1: How does the Touchless Hybrid mode work?

Answer: You can use the RVM kiosk without touching a single button! Simply open the **RVM Mobile App** and point your camera at the 'TOUCHLESS SCAN TO START' QR code on the kiosk screen. The entrance gate opens automatically. After inserting your bottles/cans, scan the on-screen wallet QR code to receive your reward points directly into your digital wallet.

Q2: Why did the machine reject my plastic bottle?

Answer: The most common cause is residual liquid inside the bottle. Optical sensors detect liquid refraction and reject the bottle to prevent sticky contamination inside the bin. Please empty all liquids completely before inserting. Rejection can also occur if the bottle is severely crushed or is an unsupported plastic type (e.g. PVC or HDPE detergent jugs).

Q3: What happens if the internet connection is temporarily offline?

Answer: Your points are 100% safe! The RVM kiosk operates a resilient local Microsoft SQL Server database. When the cloud connection is offline, points are saved locally in an offline queue and automatically synchronized to the Central Cloud server as soon as the network reconnects.

Q4: How many containers can I deposit in a single session?

Answer: There is no limit! You can insert as many bottles, cans, and cartons as you wish during one session. The live counter tracks every accepted container. When you are finished, press **Enter** to claim all points together.

Q5: How do I redeem my points for balance or gifts?

Answer: Visit the official web portal at isprvm.binishaqsoft.com or visit the campus partner desk. Enter your registered 11-digit mobile number to view balance vouchers, mobile recharge codes, and gift rewards.

Q6: How do I scan the QR code to claim my points?

Answer: Open the **RVM Mobile App** on your smartphone, tap the **Scan QR** tab, and point your camera at the kiosk screen. Your points will be credited immediately to your account. Alternatively, you can type the 6-character session code shown below the QR code into the app.